



Cape Peninsula
University of Technology

FACULTY OF INFORMATICS & DESIGN

Advanced Diploma in Information & Communication Technology

APPLICATIONS DEVELOPMENT PRACTICE 4: ADP470S

SUBJECT GUIDE:	2026
COURSE CODE:	ADICTA
NQF LEVEL:	7
PREREQUISITE SUBJECTS:	PREREQUISITE SUBJECTS: Diploma NQF Level 6

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1. INTRODUCTION

The subject *Application Development Practice* is designed to cultivate professional-level problem-solving competencies within the domain of software development. Successful completion of this course will provide students with a comprehensive understanding of fundamental principles of data representation and algorithm design, as well as the analytical skills required to address algorithmic challenges related to feasibility, efficiency, and performance in software development contexts.

The *Capita selecta* for this subject draws upon topics from discrete mathematics, data structures and algorithms, with particular emphasis on the systematic analysis and comparative evaluation of algorithmic performance.

2. GENERAL

2.1 Contact Information

DESIGNATION	NAME	LOCATION	TELEPHONE NUMBER	E-MAIL ADDRESS
Subject Co-ordinator	T. Makhurane	Engineering 2.42	(021) 460 3178	makhuranet@cput.ac.za
Lecturers	T. Makhurane	Engineering 2.42	(021) 460 3178	makhuranet@cput.ac.za
Domain Coordinator	T. Makhurane	Engineering 2.42	(021) 460 3178	makhuranet@cput.ac.za
Head of Department	T. Ncubukezi	Engineering 2.30	(021) 460-3713	ncubukezit@cput.ac.za
Secretary	N. Allie	Engineering 2.1	(021) 460-3010	alien@cput.ac.za
Support Technician				
Support Technician	R. Pietersen	Engineering 1.4	(021) 460-4209	pietersenr@cput.ac.za
Support Technician	I. Saliem	Engineering 1.28	(021) 460-3031	saliemi@cput.ac.za
Support Technician				

2.2 Class Times and Venues

As published on <http://myclassroom.cput.ac.za> and notice boards. The subject consists of one in-person classroom session every week. Additional classes may be scheduled however, as and when necessary.

2.3 Learner Management System (LMS)

<https://myclassroom.cput.ac.za> Blackboard used for general communications and announcements, posting of course material, and assignments. It is the responsibility of the learner to check Blackboard regularly.

2.4 Crims System

<https://enquiries.cput.ac.za> The crims system is an online platform to help students open tickets and raise subject and other academic related queries. Students must not open a crims ticket if the matter can be addressed with the Lecturer/Subject Coordinator directly.

3. STUDY MATERIALS

3.1 Prescribed Textbook

A number of readings will be recommended by the lecturer and included on slides and from time to time. The following may prove useful but are not essential:

- Any academic text on data structures and algorithms
- Any academic text on discrete mathematics

3.2 Recommended Readings

Material on Data Structures and Algorithms is available on many university websites. In addition, many books have been published on Data Structures and Algorithms. The lecturers will be pointing out readings of interest to the students as the subject progresses

3.3 Websites

Any websites on data structures and algorithms

Any websites on discrete mathematics

4. MODULE CREDITS

This module has 20 SAQA or 0.167 HEMIS credits on NQF level 7. Please refer to the Faculty prospectus on <http://www.cput.ac.za/academic/faculties/informaticsdesign/prospectus> for further information.

4.1 Planning of Time Allocation for Learning Activities

There are 200 notional hours allocated to this module. These notional hours comprise of 60 hours (30%) for contact time and the remainder will be self-study and other tasks, assignments, and tests.

METHOD	% OF YOUR LEARNING TIME
Seminars / Practicals (face-to-face, limited interaction or technologically mediated)	30%
Independent self-study of standard texts and references and specially prepared materials (study guides, books, journals, using online tutorials etc.)	20%
Independent self-study, time needed to prepare for assessment (e.g. tests/exams)	25%
Student learning groups (working with your peers for group assignments)	25%

5. ETHICAL CONDUCT

5.1 Roles and Responsibilities

Kindly refer to CPUT's Policy Docs and Student Guidelines:

<https://www.cput.ac.za/storage/students/2024/2024-Student-Rules-and-Regulations-V3.pdf>

5.2 Class attendance and participation

Attendance is compulsory. Kindly refer to page 19 of CPUT's General Handbook Academic and Student Rules and Regulations: <https://www.cput.ac.za/storage/students/2024/2024-Student-Rules-and-Regulations-V3.pdf>

5.3 Grievance procedures

Kindly refer to page 76 of CPUT's General Handbook Academic and Student Rules and Regulations:

<https://www.cput.ac.za/storage/students/2024/2024-Student-Rules-and-Regulations-V3.pdf>

5.4 Plagiarism

Kindly refer to page 29 of CPUT's General Handbook Academic and Student Rules and Regulations:

<https://www.cput.ac.za/storage/students/2024/2024-Student-Rules-and-Regulations-V3.pdf>

6. ASSESSMENT

6.1 Assessment policy and regulations

Assessment in this subject follows a combination of formative and summative approaches to support progressive learning and competency development. Assessment methods include written tests, assignments, exercises, and an individual project. All assessments are designed to evaluate both theoretical understanding and practical application in alignment with the stated learning outcomes.

This subject includes **three major tests**. In addition, a **major individual project** will be set as a core assessment component. Minor assessment tasks, including assignments, quizzes, and exercises, may also be administered as required to support the progressive development of relevant skills.

The major tests will be conducted as written assessments in a classroom or examination venue. Assignments will generally have a submission period of one week and must be submitted electronically via Blackboard. Project-based assessments will be allocated extended completion periods appropriate to their scope.

Assessment submissions may consist of scanned copies of handwritten work or typed documents submitted in .docx or .pdf format.

Test queries: Any queries regarding the marking or content of a test must be raised within one week of the test being reviewed in class. After this period, test marks will be considered final and will not be amended.

Absence from a test: In the event of absence from a scheduled test, students are required to submit a valid medical certificate to their lecturer (via the Secretary) within one week of the test date. In addition, the lecturer must be notified on the day of the test of the student's illness.

Due dates must be strictly adhered to. **NO late submissions are allowed for any subject in this programme.**

Assignments may be administered via Blackboard and other platforms.

6.2 Assessment Opportunities: Administration

The Table below summarises the marks weighting.

ASSESSMENT BREAKDOWN AND WEIGHTS					
ASSESSMENT CRITERIA	1 ST TERM	2 ND TERM	3 RD TERM	4 TH TERM	TOTALS
Nature of assessment	Test (T1)	Test (T2)	Test (T3)	Test (T4)	
Will assessment be included in this term?	Yes	Yes	Yes	Yes	
Total assessment weight for the term	10%	35%	20%	35%	100%
Will the assessment be moderated?	No	Yes	No	*Yes	

STUDY COMPONENT

7. SUBJECT SPECIFICATIONS

7.1 Purpose of the Subject

Topics included in this *Capita selecta* include items drawn from Data Structures and Algorithms and equally importantly the Analysis and comparison pertaining to the performance of algorithms. For implementation purposes students can make use of an object-oriented language. We will work primarily with Java. Alternatively, Python or C# may also be used. The purpose of this subject is to develop learners' professional-level problem-solving competencies in software development. The subject equips learners with foundational and intermediate knowledge of data structures, algorithms, and algorithmic analysis, enabling them to design, implement, and evaluate efficient computational solutions. Emphasis is placed on algorithmic feasibility, correctness, and performance within practical software development contexts.

Upon successful completion of this subject, learners will be able to:

- Demonstrate understanding of fundamental data structures and algorithms.

- Apply algorithmic techniques to solve computational problems efficiently.
- Analyse and compare algorithms in terms of time and space complexity.
- Implement and evaluate data structures suitable for specific application contexts.
- Apply theoretical concepts to practical problem-solving scenarios.

7.2 Subject Structure

7.2.1 Subject Outcomes

	TERM 1
SUBJECT OUTCOMES	DESCRIPTION
SO1	Basic Introduction to Algorithms and Complexity
SO2	Recursion versus Iteration, example of Factorial and Fibonacci
SO3	Idea about complexity of an Algorithm with the help of Linear Search, and Binary Search
SO4	Bubble Sort and Best-case, worst case and average case complexity using bubble sort Algorithm example
SO5	2d -and Multidimensional Arrays. Introduction to Linear Linked list with discussion on limitation of an Array
SO6	Limitation of single Linked List and introduction to Doubly and Circular Linked List
SO7	Hand out exercises on all the topics discussed in Term 1. Exercises will be applications of the theory discussed.
Assessment	T1 Written Assessment (Test) (10%)
	TERM 2
SUBJECT OUTCOMES	DESCRIPTION
SO1	Review of the 10 exercises given in T1

SO2	Exercises Review
SO3	Re-visit recursion and discuss Stack operations
SO4	Q operations, Circular Q
SO5	Priority Queue
SO6	Binary trees, Traversals
SO7	Introduction to AVL / Height Balanced Binary Tree
SO8	Short test
SO9	Sorting Algorithms and Algorithm Analysis
S10	Sorting Algorithms and Algorithm Analysis continues
S11	Sorting Algorithms and Algorithm Analysis continues
S12	Sorting Algorithms and Algorithm Analysis continues
S13	Sorting Algorithms and Algorithm Analysis
Assessment	T2 Written Assessment (Test)

7.2.2 Subject Breakdown Schedule

PUBLIC HOLIDAYS AND OTHER IMPORTANT DATES:

- 16 Jun – Youth Day

	TERM 3			
SUBJECT OUTCOMES	THEMES/SPECIFIC OUTCOMES			
SO1	Introduction to Graphs and representation of graphs in computer			
SO2	Graph traversal Algorithms			
SO3	Continue Graph Traversal Algorithms			
SO4	Short test			
SO5	Shortest path Algorithms (Prims, Kruskal, Dijkstra, Floyd)			
SO6	Shortest path Algorithms (Disjoint Set, Kruskal, Floyd)			
SO7	Hashing (BB announcement for extended class)			

S08	
Assessment	Project T3
	TERM 4
SUBJECT OUTCOMES	THEMES/SPECIFIC OUTCOMES
S01	GPS and bitmap image data Structures
S02	Exercises on week 1 discussion
S03	Parity bit / Even – odd parity / CRC with coding
S04	
	Review and Clarification
S05	Assessment
S06	Students who missed any assessments for a University Approved reason contacts the Lecturer(s) TA and appear for the same (Sick Test week)
Assessment	T4 Written Assessment

7.3 Articulation with other subjects in the programme

This subject is not directly linked to a subject in Post Graduate Diploma.

7.4 Learning presumed to be in place

Diploma or any other equivalent NQF level 6 qualification.

7.5 Critical Cross-field Outcomes

CRITICAL CROSS-FIELD OUTCOMES	DESCRIPTION
CCFO1	Identify and solve problems with responses which display responsible decisions, using critical thinking, have been made.
CCFO2	Work effectively with others as a member of a team, group, organisation or community.
CCFO3	Organise and manage oneself and one's activities responsibly and effectively.
CCFO4	Collect, analyse, organise and critically evaluate information.
CCFO5	Communicate effectively, using visual, mathematical, and/or language skills in the modes of written and/or oral presentation.
CCFO6	Use science and technology effectively and critically, showing responsibility towards the environment and the health of others.
CCFO7	Demonstrate an understanding of the world as a set of related systems by recognising that problem-solving contexts do not exist in isolation.
CCFO8	Contribute to the full personal development of each learner and the social and economic development of the society at large, by making it the underlying intention of any programme of learning to create awareness in individuals about the importance of: i) reflecting on and exploring a variety of strategies to learn more effectively; ii) participating as responsible citizens in the life of local, national and global communities; and iii) being culturally and aesthetically sensitive across a range of social contexts. (RSASAQA, 2001:24)

8. STUDY UNITS/STUDY THEMES

8.1 Study Units

Refer to 7.2.2 Subject Breakdown Structure.

8.2 Assessment Opportunities and Assessment strategy

Refer to 6.2 Assessment Opportunities: Administration for the linkages. Furthermore, a comprehensive rubric will be published on Blackboard for each assignment and project work.

8.3 Subject Outcomes and Associated Assessment Criteria

Refer to 7.2.2 Subject Breakdown Structure for the linkages. Furthermore, a comprehensive rubric will be published on Blackboard for each assignment and project work.

8.4 Self-study Activities

Exercises and quizzes.

9. STUDENT ACKNOWLEDGEMENT

CAPE PENINSULA UNIVERSITY OF TECHNOLOGY STUDENT ACKNOWLEDGEMENT

2026

ADVANCED DIPLOMA in INFORMATION and COMMUNICATION TECHNOLOGY: APPLICATIONS DEVELOPMENT (ADICTA) APPLICATION DEVELOPMENT PRACTICE 4 (ADP470S)

ACKNOWLEDGE THAT YOU HAVE READ THE SUBJECT GUIDE AND UNDERSTAND ITS CONTENT

Please read the following statement and complete the spaces that follow. Make a copy of this page for your own reference. Hand-in this original page during the next class session.

I have read the subject guide of **APPLICATION DEVELOPMENT PRACTICE 4** and understand the information and my responsibilities as a learner for this subject. I am also aware that the Subject Guide may be revised due to unforeseen circumstances and it remains my responsibility to adhere to the most recent guide.

Name & Surname:
(Print)

Student Number:

Contact number: (home/res) Contact number: (cell)

Signature: Date: